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Hermann Haken

Synergetics

An Introduction

Nonequilibrium Phase Transitions and Self-Organization
in Physics, Chemistry, and Biology

Third Revised and Enlarged Edition

With 161 Figures

Springer-Verlag
Berlin Heidelberg New York Tokyo 1983

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ISBN-13: 978-3-642-88340-8 e-ISBN-13: 978-3-642-88340-8
DOI: 10.1007/978-3-642-88340-8

Library of Congress Cataloging in Publication Data. Haken, H. Synergetics : an introduction : nonequilibrium phase transitions and self-organization in physics, chemistry. (Springer series in synergetics ; v. 1). 1. System theory. 2. Self-organizing systems. I. Title. II. Series. Q295.H35 1983 003 83-4682

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Softcover reprint of the hardcover 3rd 1983

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Satz: Daten- und Lichtsatz-Service, Würzburg

Offset printing and bookbinding: Konrad Triltsch, Graphischer Betrieb, Würzburg

2153/3130 – 543210

To the Memory of

Maria and Anton Vollath

Preface to the Third Edition

Over the past years the field of synergetics has been mushrooming. An ever-increasing number of scientific papers are published on the subject, and numerous conferences all over the world are devoted to it. Depending on the particular aspects of synergetics being treated, these conferences can have such varied titles as “Nonequilibrium Nonlinear Statistical Physics,” “Self-Organization,” “Chaos and Order,” and others. Many professors and students have expressed the view that the present book provides a good introduction to this new field. This is also reflected by the fact that it has been translated into Russian, Japanese, Chinese, German, and other languages, and that the second edition has also sold out. I am taking the third edition as an opportunity to cover some important recent developments and to make the book still more readable.

First, I have largely revised the section on self-organization in continuously extended media and entirely rewritten the section on the Bénard instability. Second, because the methods of synergetics are penetrating such fields as economics, I have included an economic model on the transition from full employment to underemployment in which I use the concept of nonequilibrium phase transitions developed elsewhere in the book. Third, because a great many papers are currently devoted to the fascinating problem of chaotic motion, I have added a section on discrete maps. These maps are widely used in such problems, and can reveal period-doubling bifurcations, intermittency, and chaos.

Instability hierarchies, which are dealt with in this book in the context of the laser, have proved since its writing to be a widespread phenomenon in systems driven far from thermal equilibrium. Because a general treatment of this problem would have gone beyond the scope and level of the present monograph, however, I have written a further book entitled *Advanced Synergetics**, which can be considered a continuation of the present text.

I wish to thank my co-worker Dr. A. Wunderlin for his assistance in revising this book, and especially for his efforts in the Bénard problem. I am also greatly indebted to my secretary, Mrs. U. Funke, for her efficient help in reworking several chapters.

Stuttgart, January 1983

Hermann Haken

* Springer Ser. Synergetics, Vol. 20 (Springer, Berlin, Heidelberg, New York, Tokyo 1983)

Preface to the Second Edition

The publication of this second edition was motivated by several facts. First of all, the first edition had been sold out in less than one year. It had found excellent critics and enthusiastic responses from professors and students welcoming this new interdisciplinary approach. This appreciation is reflected by the fact that the book is presently translated into Russian and Japanese also.

I have used this opportunity to include some of the most interesting recent developments. Therefore I have added a whole new chapter on the fascinating and rapidly growing field of chaos dealing with irregular motion caused by deterministic forces. This kind of phenomenon is presently found in quite diverse fields ranging from physics to biology. Furthermore I have included a section on the analytical treatment of a morphogenetic model using the order parameter concept developed in this book. Among the further additions, there is now a complete description of the onset of ultrashort laser pulses. It goes without saying that the few minor misprints or errors of the first edition have been corrected.

I wish to thank all who have helped me to incorporate these additions.

Stuttgart, July 1978

Hermann Haken

Preface to the First Edition

The spontaneous formation of well organized structures out of germs or even out of chaos is one of the most fascinating phenomena and most challenging problems scientists are confronted with. Such phenomena are an experience of our daily life when we observe the growth of plants and animals. Thinking of much larger time scales, scientists are led into the problems of evolution, and, ultimately, of the origin of living matter. When we try to explain or understand in some sense these extremely complex biological phenomena, it is a natural question whether processes of self-organization may be found in much simpler systems of the unanimated world.

In recent years it has become more and more evident that there exist numerous examples in physical and chemical systems where well organized spatial, temporal, or spatio-temporal structures arise out of chaotic states. Furthermore, as in living organisms, the functioning of these systems can be maintained only by a flux of energy (and matter) through them. In contrast to man-made machines, which are devised to exhibit special structures and functionings, these structures develop spontaneously—they are self-organizing. It came as a surprise to many

scientists that numerous such systems show striking similarities in their behavior when passing from the disordered to the ordered state. This strongly indicates that the functioning of such systems obeys the same basic principles. In our book we wish to explain such basic principles and underlying conceptions and to present the mathematical tools to cope with them.

This book is meant as a text for students of physics, chemistry and biology who want to learn about these principles and methods. I have tried to present mathematics in an elementary fashion wherever possible. Therefore the knowledge on an undergraduate course in calculus should be sufficient. A good deal of important mathematical results is nowadays buried under a complicated nomenclature. I have avoided it as far as possible through, of course, a certain number of technical expressions must be used. I explain them wherever they are introduced. Incidentally, a good many of the methods can also be used for other problems, not only for self-organizing systems. To achieve a self-contained text I included some chapters which require some more patience or a more profound mathematical background of the reader. Those chapters are marked by an asterisk. Some of them contain very recent results so that they may also be profitable for research workers.

The basic knowledge required for the physical, chemical and biological systems is, on the average, not very special. The corresponding chapters are arranged in such a way that a student of one of these disciplines need only to read "his" chapter. Nevertheless it is highly recommended to browse through the other chapters just to get a feeling a how analogous all these systems are among each other. I have called this discipline "synergetics". What we investigate is the joint action of many subsystems (mostly of the same or of few different kinds) so as to produce structure and functioning on a macroscopic scale. On the other hand, many different disciplines cooperate here to find general principles governing self-organizing systems.

I wish to thank Dr. Lotsch of Springer-Verlag who suggested writing an extended version of my article "Cooperative phenomena in systems far from thermal equilibrium and in nonphysical systems", in *Rev. Mod. Phys.* (1975). In the course of writing the "extension", eventually a completely new manuscript evolved. I wanted to make this field especially understandable to students of physics, chemistry and biology. In a way, this book and my previous article have become complementary.

It is a pleasure to thank my colleagues and friends, especially Prof. W. Weidlich, for many fruitful discussions over the years. The assistance of my secretary, Mrs. U. Funke, and of my coworker Dr. A. Wunderlin was an enormous help for me in writing this book and I wish to express my deep gratitude to them. Dr. Wunderlin checked the formulas very carefully, recalculating many of them, prepared many of the figures, and made valuable suggestions how to improve the manuscript. In spite of her extended administrative work, Mrs. U. Funke has drawn most of the figures and wrote several versions of the manuscript, including the formulas, in a perfect way. Her willingness and tireless efforts encouraged me again and again to complete this book.

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